LAB 6

Code and Branch Coverage Analysis

Marathon Registration

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Objective

Measure and improve line and branch coverage for the marathon registration application using pytest and coverage.py.

Results

LINE COVERAGE	BRANCH COVERAGE	TEST RESULTS
100%	100%	13 line / 34 branch

Approach

The line-coverage suite exercises the application's functions and return paths. The branch suite adds cases for fee decisions, age limits, wave thresholds, race limits, and missed checkpoints.

Terminal evidence follows; separate test files contain the test suites. Coverage shows which code was exercised, but does not by itself prove that every possible input is correct.

Line coverage

13 TESTS PASSED / 38 STATEMENTS COVERED

Final terminal output

```
PS D:\Documents\Sem-5\SE\SE-Labs-PES1UG24CS123\Lab6\L6_L7_60_marathon_registration\LAB6> .\.venv\Scripts\python.exe
-m coverage run --source=marathon_registration -m pytest -v CodeCoverageTest_60_marathon_registration.py
rootdir: D:\Documents\Sem-5\SE\SE-Labs-PES1UG24CS123\Lab6\L6_L7_60_marathon_registration\LAB6
collected 13 items

CodeCoverageTest_60_marathon_registration.py::test_entry_fee_rejections[ultra-30-unknown race] PASSED        [  7%]
CodeCoverageTest_60_marathon_registration.py::test_entry_fee_rejections[5k-11-too young] PASSED             [ 15%]
CodeCoverageTest_60_marathon_registration.py::test_entry_fee_rejections[half-17-too young for this distance] PASSED
[ 23%]
CodeCoverageTest_60_marathon_registration.py::test_entry_fee_discounts_and_tshirt PASSED                    [ 30%]
CodeCoverageTest_60_marathon_registration.py::test_waves[None-full-D] PASSED                                [ 38%]
CodeCoverageTest_60_marathon_registration.py::test_waves[180-full-A] PASSED                                 [ 46%]
CodeCoverageTest_60_marathon_registration.py::test_waves[240-full-B] PASSED                                 [ 53%]
CodeCoverageTest_60_marathon_registration.py::test_waves[241-full-C] PASSED                                 [ 61%]
CodeCoverageTest_60_marathon_registration.py::test_waves[90-5k-A] PASSED                                    [ 69%]
CodeCoverageTest_60_marathon_registration.py::test_status[100-5k-1-DSQ] PASSED                              [ 76%]
CodeCoverageTest_60_marathon_registration.py::test_status[360-full-0-finisher] PASSED                       [ 84%]
CodeCoverageTest_60_marathon_registration.py::test_status[180-half-0-finisher] PASSED                       [ 92%]
CodeCoverageTest_60_marathon_registration.py::test_status[91-5k-0-DNF] PASSED                               [100%]

============================================== 13 passed in 0.28s ==============================================
PS D:\Documents\Sem-5\SE\SE-Labs-PES1UG24CS123\Lab6\L6_L7_60_marathon_registration\LAB6> .\.venv\Scripts\python.exe
-m coverage report -m
Name                       Stmts   Miss  Cover   Missing
--------------------------------------------------------
marathon_registration.py      38      0   100%
--------------------------------------------------------
TOTAL                         38      0   100%
PS D:\Documents\Sem-5\SE\SE-Labs-PES1UG24CS123\Lab6\L6_L7_60_marathon_registration\LAB6>
```

Interpretation

The report shows 38 statements, 0 missed and 100% line coverage for **marathon_registration.py**. All 13 cases in the line-coverage test file passed.

Test file

CodeCoverageTest_60_marathon_registration.py

Branch coverage

34 TESTS PASSED / 28 OF 28 BRANCHES COVERED

Test execution

```
PS D:\Documents\Sem-5\SE\SE-Labs-PES1UG24CS123\Lab6\L6_L7_60_marathon_registration\LAB6> .\.venv\Scripts\python.exe
BranchCoverageTest_60_marathon_registration.py::test_waves[90-5k-A] PASSED                                  [ 26%]
BranchCoverageTest_60_marathon_registration.py::test_status[100-5k-1-DSQ] PASSED                            [ 29%]
BranchCoverageTest_60_marathon_registration.py::test_status[360-full-0-finisher] PASSED                     [ 32%]
BranchCoverageTest_60_marathon_registration.py::test_status[180-half-0-finisher] PASSED                     [ 35%]
BranchCoverageTest_60_marathon_registration.py::test_status[91-5k-0-DNF] PASSED                             [ 38%]
BranchCoverageTest_60_marathon_registration.py::test_fee_boundaries[10k-30-False-False-800] PASSED          [ 41%]
BranchCoverageTest_60_marathon_registration.py::test_fee_boundaries[5k-12-False-False-500] PASSED           [ 44%]
BranchCoverageTest_60_marathon_registration.py::test_fee_boundaries[5k-64-False-False-500] PASSED           [ 47%]
BranchCoverageTest_60_marathon_registration.py::test_fee_boundaries[5k-65-False-False-250] PASSED           [ 50%]
BranchCoverageTest_60_marathon_registration.py::test_fee_boundaries[half-18-False-False-1200] PASSED        [ 52%]
BranchCoverageTest_60_marathon_registration.py::test_fee_boundaries[full-18-False-False-1800] PASSED        [ 55%]
BranchCoverageTest_60_marathon_registration.py::test_wave_boundaries[90-5k-A] PASSED                        [ 58%]
BranchCoverageTest_60_marathon_registration.py::test_wave_boundaries[91-5k-B] PASSED                        [ 61%]
BranchCoverageTest_60_marathon_registration.py::test_wave_boundaries[120-5k-B] PASSED                       [ 64%]
BranchCoverageTest_60_marathon_registration.py::test_wave_boundaries[121-5k-C] PASSED                       [ 67%]
BranchCoverageTest_60_marathon_registration.py::test_wave_boundaries[180-full-A] PASSED                     [ 70%]
BranchCoverageTest_60_marathon_registration.py::test_wave_boundaries[181-full-B] PASSED                     [ 73%]
BranchCoverageTest_60_marathon_registration.py::test_wave_boundaries[240-full-B] PASSED                     [ 76%]
BranchCoverageTest_60_marathon_registration.py::test_wave_boundaries[241-full-C] PASSED                     [ 79%]
BranchCoverageTest_60_marathon_registration.py::test_status_boundaries[90-5k-0-finisher] PASSED             [ 82%]
BranchCoverageTest_60_marathon_registration.py::test_status_boundaries[91-5k-0-DNF] PASSED                  [ 85%]
BranchCoverageTest_60_marathon_registration.py::test_status_boundaries[180-half-0-finisher] PASSED          [ 88%]
BranchCoverageTest_60_marathon_registration.py::test_status_boundaries[181-half-0-DNF] PASSED               [ 91%]
BranchCoverageTest_60_marathon_registration.py::test_status_boundaries[360-full-0-finisher] PASSED          [ 94%]
BranchCoverageTest_60_marathon_registration.py::test_status_boundaries[361-full-0-DNF] PASSED               [ 97%]
BranchCoverageTest_60_marathon_registration.py::test_status_boundaries[1-full-1-DSQ] PASSED                 [100%]

============================================== 34 passed in 0.62s ==============================================
PS D:\Documents\Sem-5\SE\SE-Labs-PES1UG24CS123\Lab6\L6_L7_60_marathon_registration\LAB6> .\.venv\Scripts\python.exe
-m coverage json -o branch_coverage.json
Wrote JSON report to branch_coverage.json
PS D:\Documents\Sem-5\SE\SE-Labs-PES1UG24CS123\Lab6\L6_L7_60_marathon_registration\LAB6> .\.venv\Scripts\python.exe
-m coverage json -o branch_coverage.json
```

Final branch coverage report

```
PS D:\Documents\Sem-5\SE\SE-Labs-PES1UG24CS123\Lab6\L6_L7_60_marathon_registration\LAB6> .\.venv\Scripts\python.exe
-m coverage report -m
Name                       Stmts   Miss Branch BrPart  Cover   Missing
----------------------------------------------------------------------
marathon_registration.py      38      0     28      0   100%
----------------------------------------------------------------------
TOTAL                         38      0     28      0   100%
PS D:\Documents\Sem-5\SE\SE-Labs-PES1UG24CS123\Lab6\L6_L7_60_marathon_registration\LAB6> .\.venv\Scripts\python.exe
-c "import json; t=json.load(open('branch_coverage.json'))['totals']; print('Branch coverage: {} / {} = {:.1f}%'.for
mat(t['covered_branches'], t['num_branches'], 100*t['covered_branches']/t['num_branches']))"
Branch coverage: 28 / 28 = 100.0%
PS D:\Documents\Sem-5\SE\SE-Labs-PES1UG24CS123\Lab6\L6_L7_60_marathon_registration\LAB6>
```

28 / 28 branches covered • 0 partial branches

Line-coverage test file

CodeCoverageTest_60_marathon_registration.py | COMPLETE FILE

```
import pytest
from marathon_registration import entry_fee, start_wave, finisher_status


@pytest.mark.parametrize("race, age, message", [
    ("ultra", 30, "unknown race"),
    ("5k", 11, "too young"),
    ("half", 17, "too young for this distance"),
])
def test_entry_fee_rejections(race, age, message):
    with pytest.raises(ValueError, match=message):
        entry_fee(race, age, False, False)


def test_entry_fee_discounts_and_tshirt():
    # 1800 - 25% = 1350; senior discount gives 675; T-shirt adds 300.
    assert entry_fee("full", 65, True, True) == 975


@pytest.mark.parametrize("time, race, expected", [
    (None, "full", "D"),
    (180, "full", "A"),
    (240, "full", "B"),
    (241, "full", "C"),
    (90, "5k", "A"),
])
def test_waves(time, race, expected):
    assert start_wave(time, race) == expected


@pytest.mark.parametrize("time, race, missed, expected", [
    (100, "5k", 1, "DSQ"),
    (360, "full", 0, "finisher"),
    (180, "half", 0, "finisher"),
    (91, "5k", 0, "DNF"),
])
def test_status(time, race, missed, expected):
    assert finisher_status(time, race, missed) == expected
```

Branch-coverage test file

BranchCoverageTest_60_marathon_registration.py | PART 1 OF 2

```python
import pytest
from marathon_registration import entry_fee, start_wave, finisher_status


@pytest.mark.parametrize("race, age, message", [
    ("ultra", 30, "unknown race"),
    ("5k", 11, "too young"),
    ("half", 17, "too young for this distance"),
])
def test_entry_fee_rejections(race, age, message):
    with pytest.raises(ValueError, match=message):
        entry_fee(race, age, False, False)


def test_entry_fee_discounts_and_tshirt():
    # 1800 - 25% = 1350; senior discount gives 675; T-shirt adds 300.
    assert entry_fee("full", 65, True, True) == 975


@pytest.mark.parametrize("time, race, expected", [
    (None, "full", "D"),
    (180, "full", "A"),
    (240, "full", "B"),
    (241, "full", "C"),
    (90, "5k", "A"),
])
def test_waves(time, race, expected):
    assert start_wave(time, race) == expected


@pytest.mark.parametrize("time, race, missed, expected", [
    (100, "5k", 1, "DSQ"),
    (360, "full", 0, "finisher"),
    (180, "half", 0, "finisher"),
    (91, "5k", 0, "DNF"),
])
def test_status(time, race, missed, expected):
    assert finisher_status(time, race, missed) == expected
@pytest.mark.parametrize("race, age, early, shirt, expected", [
    ("10k", 30, False, False, 800),
```

Branch-coverage test file

BranchCoverageTest_60_marathon_registration.py | PART 2 OF 2

```
    ("5k", 12, False, False, 500),
    ("5k", 64, False, False, 500),
    ("5k", 65, False, False, 250),
    ("half", 18, False, False, 1200),
    ("full", 18, False, False, 1800),
])
def test_fee_boundaries(race, age, early, shirt, expected):
    assert entry_fee(race, age, early, shirt) == expected


@pytest.mark.parametrize("time, race, expected", [
    (90, "5k", "A"),
    (91, "5k", "B"),
    (120, "5k", "B"),
    (121, "5k", "C"),
    (180, "full", "A"),
    (181, "full", "B"),
    (240, "full", "B"),
    (241, "full", "C"),
])
def test_wave_boundaries(time, race, expected):
    assert start_wave(time, race) == expected


@pytest.mark.parametrize("time, race, missed, expected", [
    (90, "5k", 0, "finisher"),
    (91, "5k", 0, "DNF"),
    (180, "half", 0, "finisher"),
    (181, "half", 0, "DNF"),
    (360, "full", 0, "finisher"),
    (361, "full", 0, "DNF"),
    (1, "full", 1, "DSQ"),
])
def test_status_boundaries(time, race, missed, expected):
    assert finisher_status(time, race, missed) == expected
```